UI/UX Guidelines & Design System

Project: NutriVision — AI-Driven Dietary Health Assistant

Official Documentation

1. Design Principles

The core philosophy of NutriVision revolves around simplicity, encouragement, and seamless integration into the user's daily routine.

- **Mobile-First & Ergonomic:** Core actions like scanning and logging are positioned at the bottom for easy one-handed reach.
- **Frictionless Interaction:** AI-driven scanning reduces manual data entry, turning a chore into a quick, satisfying action.
- **Encouraging Aesthetic:** The UI acts as a supportive coach, using soft rounded shapes and uncluttered spaces to reduce cognitive load.

2. Color Schemes

The color system emphasizes health, organic growth, and clarity.

- **Primary Brand Green:** Used for primary call-to-action (CTA) buttons, the active state of the central "Scan" navigation item, and the splash screen background.
- **Dark Green/Teal:** Applied to high-level typography, including page titles, primary statistics, and the background of the camera viewfinder.
- **Background Colors:** The application utilizes an off-white/light gray for the main app background to reduce eye strain, while utilizing pure white for content cards and input fields to create depth.
- **Semantic & Accent Colors:** Orange is utilized specifically to denote "Fats" in macro tracking, while lighter, desaturated greens represent incomplete progress or inactive chart states.

3. Typography

The application uses a clean, modern sans-serif typeface to ensure readability of data-heavy interfaces.

- **Headings (H1/H2):** Bold and heavily weighted, utilizing the Dark Green color (e.g., "Your profile", "Good to see you again.").
- **Body Text:** Regular weight, often utilizing lighter gray colors for subtitles and helper text (e.g., "Smarter nutrition. Healthier you.").
- **Numeric Data:** Highlighted using bold, scaled-up fonts for primary metrics like remaining calories, ensuring instant scannability.

4. UI Components & Layout

The interface is structured around modular, rounded components that create a soft and approachable aesthetic.

- **Cards & Containers:** Content is grouped into white cards with soft rounded corners (approximately 16px–24px radius) and very subtle borders or drop shadows.
- **Buttons:** Primary CTA buttons (like "Create account" and "Take a photo") span the full width of their containers with highly rounded, pill-shaped edges.
- **Bottom Navigation:** Features five distinct tabs. Inactive icons are outlined in gray, while the active tab is highlighted with a green icon and text. The central "Scan" action is emphasized with a distinct solid green background container.
- **Input Fields:** Form fields are bordered with rounded corners and utilize left-aligned leading icons (user, email, lock) to guide user input.

5. Data Visualization

Nutritional data is presented using simple, glanceable graphics rather than complex spreadsheets.

- **Circular Progress:** The daily calorie intake is represented by a segmented circular ring on the Home dashboard.
- **Linear Progress Bars:** Macronutrients (Protein, Carbs, Fats) are displayed using horizontal, color-coded progress bars.
- **Bar Charts:** Weekly caloric trends are visualized using vertical, rounded bar charts, with the current day highlighted in the primary green.

6. UX & Copywriting Principles

The user experience is designed to be frictionless, encouraging, and highly personalized.

- **Frictionless Logging:** The AI meal scanner acts as the core feature, minimizing the manual data entry typically required in diet apps.
- **Motivational Microcopy:** The app utilizes an encouraging, conversational tone, employing phrases like "Make room for better habits," "Looking good, Alex," and "Keep it going" to maintain user motivation.

7. Accessibility Considerations

NutriVision ensures inclusive design practices so the app is usable by everyone, regardless of physical or situational limitations.

- **Touch Targets:** All primary interactive buttons and navigation tabs have a minimum touch area of 48x48dp to ensure usability on smaller mobile screens.
- **Color Contrast & Legibility:** The use of deep Dark Green text on Off-White backgrounds maintains high contrast ratios (exceeding WCAG AA standards) for optimal readability.
- **Clear Visual Feedback:** Distinct color changes (e.g., active vs. inactive navigation icons) and clear progress bars provide immediate system status without relying solely on text.